# NATURE-BIOINSPIRED MATERIALS AND MECHANISMS

## Bio Inspired Materials

These are synthetic materials whose structure, properties or function mimic those of natural materials or living matter.

Examples:

- Light-harvesting photonic materials that mimic photosynthesis
- Structural composites that imitate the structure of nacre
- Metal actuators inspired by the movements of jellyfish.
- Micro fluidics and paper diagnostics
- Soft robotics.

## Echolocation

- Donald Griffin discovered bats' use of echolocation in 1940, opening what he once called a "magic well" from which scientists have been extracting knowledge ever since.
- Echolocation occurs when an animal emits a sound wave that bounces off an object, returning an echo that provides information about the object's distance and size. It is considered as nature's own SONAR system.
- Echolocation can be defined as a physiological process for locating distant or invisible objects (such as prey) by means of sound waves reflected back to the emitter (such as a bat) by the objects.
- In nature, over a thousand species echolocate, including most bats, all toothed whales, and small mammals. Many are nocturnal, burrowing, and ocean-dwelling animals that rely on echolocation to find food in an environment with little to no light. Animals have several methods for echolocation, from vibrating their throats to flapping their wings.

Examples:

(1) Bats produce high-frequency sounds to hunt for their prey.

(2) Dolphins and other toothed whales use sound waves to locate their prey and navigate their surroundings.

(3) Nocturnal oil birds use echolocation to navigate their surrounding, especially those with poor lighting conditions.

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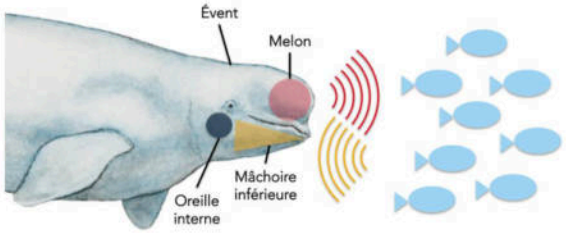
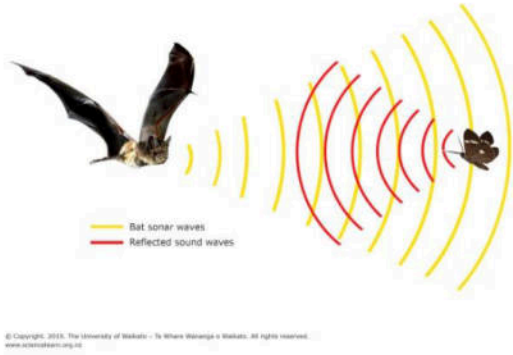
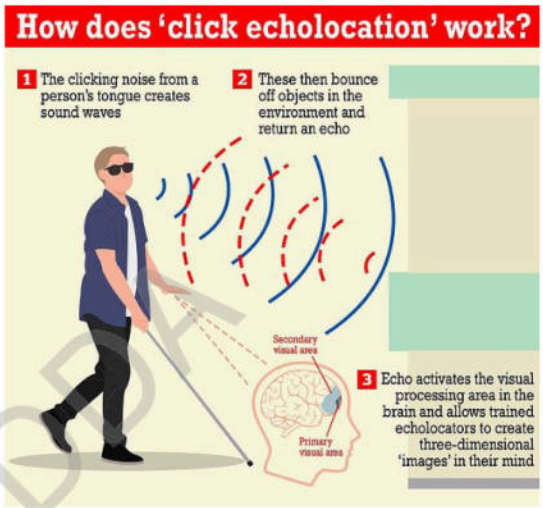



- While animals like bats and dolphins have specific sounds that they use for echolocating, humans can pick whatever sound they want to use as their sonar emission. Finger snaps, mouth clicks, and humming are some of the most common echolocating noises. Blind people also often use short and quick cane taps to echolocate.




- Echolocation calls can be composed of two different types of frequency structure: frequency modulated (FM) sweeps, and constant frequency (CF) tones. A particular call can consist of one, the other, or both structures.

**Uses**

- Echolocation, a physiological process for locating distant or invisible objects (such as prey) by means of sound waves reflected back to the emitter (such as a bat) by the objects.
- Echolocation is used for orientation, obstacle avoidance, food procurement, and social interactions.

**(i) Ultrasonography**

- A technique using echoes of ultrasound pulses to delineate objects or areas of different density in the body.
- Ian Donald was an obstetrician who developed ultrasound diagnostics during the twentieth century in Europe.

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- Doctors use ultrasound to a non-invasive imaging test, which detects the changes in the appearance of organs, tissues, and vessels and to detect abnormal masses, such as tumors.
- In an ultrasound exam, a transducer both sends the sound waves and records the echoing (returning) waves.
- The sound waves make echoes that form pictures of the tissues and organs on a computer screen (sonogram).
- Also known as sonography or diagnostic medical sonography, ultrasound imaging uses a small transducer (probe) to both transmit sound waves into the body and record the waves that echo back. Sound waves travel into the area being examined until they hit a boundary between tissues, such as between fluid and soft tissue, or soft tissue and bone.

- It allows your health care provider to see into your body without surgery. Doctors commonly use ultrasound to study a developing foetus (unborn baby), a person's abdominal and pelvic organs, muscles and tendons, or their heart and blood vessels.

The 3 main types are

- External ultrasound scan – the probe is moved over the skin.
- Internal ultrasound scan – the probe is inserted into the body.
- Endoscopic ultrasound scan – the probe is attached to a long, thin, flexible tube (an endoscope) and passed further into the body.

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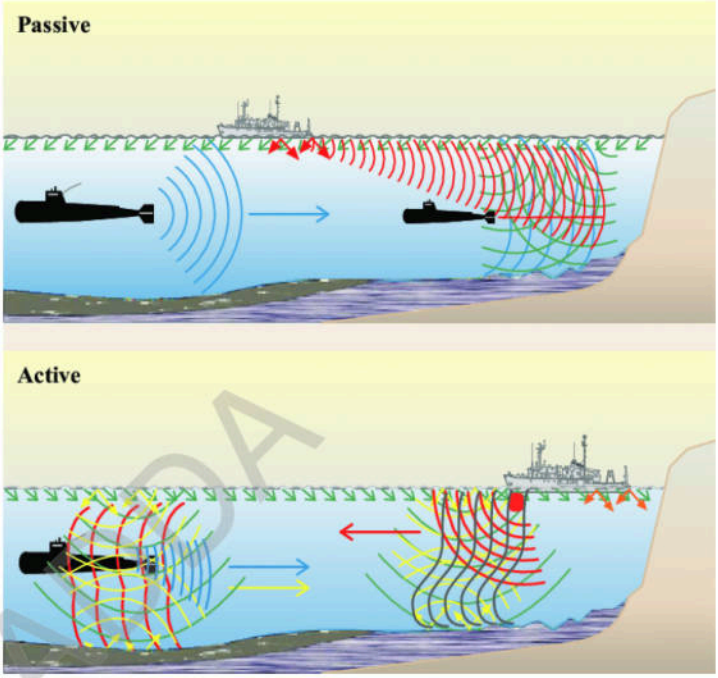
### (ii) SONAR- Sound Navigation and Ranging

- Sonar is a technique that uses sound propagation to navigate, measure distances, communicate with or detect objects on or under the surface of the water, such as other vessels.
- Reginald Fessenden discovered the SONAR technique.
- Ultrasonic waves are used in SONAR instead of audible sound waves.
- Sonar was developed during World War I as an aid in finding both submarines and icebergs. Major improvements were made on this technology during World War II, and eventually scientists adapted the highly sensitive equipment for use in oceanographic research. There are two types of sonar: active and passive.

### a) Active Sonar

- Active sonar transducers emit an acoustic signal or pulse of sound into the water. If an object is in the path of the sound pulse, the sound bounces off the object and returns an "echo" to the sonar transducer.
- If the transducer is equipped with the ability to receive signals, it measures the strength of the signal.
- By determining the time between the emission of the sound pulse and its reception, the transducer can determine the range and orientation of the object.




### b) Passive Sonar

- Passive sonar systems are used primarily to detect noise from marine objects (such as submarines or ships) and marine animals like whales.
- Unlike active sonar, passive sonar does not emit its own signal, which is an advantage for military vessels that do not want to be found or for scientific missions that concentrate on quietly "listening" to the ocean. Rather, it only detects sound waves coming towards it.
- Passive sonar cannot measure the range of an object unless it is used in conjunction with other passive listening devices. Multiple passive sonar devices may allow for triangulation of a sound source.

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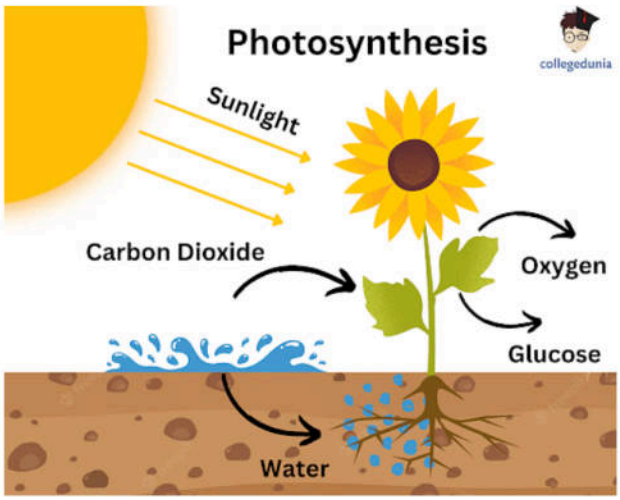
## Photosynthesis

- Photosynthesis is the process by which plants use sunlight, water, and carbon dioxide to create oxygen and energy in the form of sugar.

$$6\ CO_2 + 6\ H_2O \rightarrow C_6H_{12}O_6 + 6O_2.$$

- Photosynthesis is an endothermic reaction. This means it cannot occur without energy (from the Sun). The light required is absorbed by a green pigment called chlorophyll in the leaves.




- Chlorophyll is found in chloroplasts in plant cells, particularly the palisade and spongy mesophyll cells. The plant leaves are green because that color is the part of sunlight reflected by a pigment in the leaves called chlorophyll.
- It could also be viewed as a series of biochemical reactions that photosynthetic autotrophs *(wavelengths range from 350 to 800 nanometers)* use to convert the olar energy into potential energy, and it involves two stages.

(i) In the first stage, called *light-dependent reactions*, autotrophs capture energy from sunlight. And in the second stage, *light-independent or dark reactions*, the captured solar energy is converted into potential energy, which exists in autotrophs as chemical bonds in macromolecules.

(ii) Most of the macromolecules synthesized from photosynthesis are carbohydrates, but autotrophs can also produce proteins and lipids. These macromolecules are consumed by the autotrophs in their metabolic reactions and utilized by *heterotrophs* as food and other non-food products.

- Eg for *autotrophs*- which are organisms that are capable of producing their own food. They include archaea, prokaryotes such as purple bacteria and cyanobacteria, and eukaryotes such as algae and land plants.
- Two predominant chlorophyll species are found in plants and algae-

a) *Chlorophyll a* absorbs the most light at 372 and 642 nanometers, corresponding to violet-blue and orange lights, respectively

b) *Chlorophyll b* absorbs the most light at 392 and 626 nanometers, corresponding to blue and red lights, respectively.

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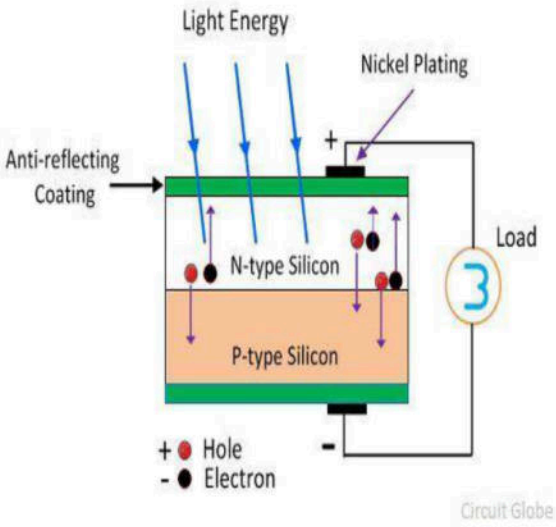
**Photovoltaic cells**

- A solar cell, or photovoltaic cell, is an electronic device that converts the energy of light directly into electricity by the photovoltaic effect, which is a physical and chemical phenomenon.
- The principle of PV effects is described such that an electric current occurs when electrons are displaced. For this to happen, photons (light particles) excite the outermost electrons of the atoms of certain semiconductor elements.




- When light radiation (E=hγ> Barrier Potential) falls on the p-n junction diode, electron-hole pairs are generated by the absorption of the radiation (Photo electric effect). These electrons gain energy and are drifted to and collected at the n-type end and the holes are drifted to and collected at the p-type end. When these two ends are electrically connected through a conductor, there is a flow of current between the two ends through the external circuit. Thus photoelectric current is produced and made available for use.

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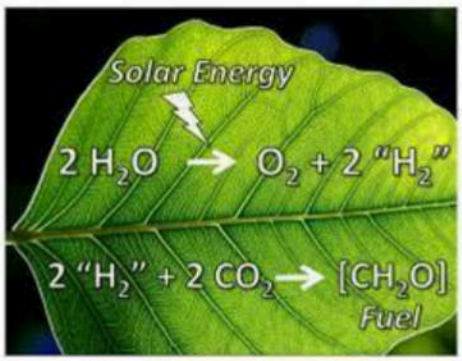
**Bionic Leaf**

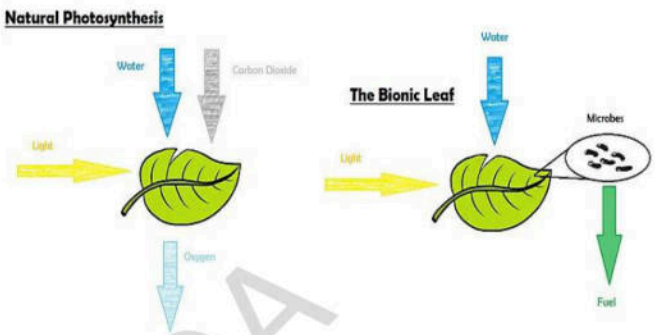
- The Bionic Leaf is a biomimetic system that gathers solar energy via photovoltaic cells that can be stored or used in a number of different functions.
- Bionic leaves can be composed of both synthetic (metals, ceramics, polymers, etc.) and organic materials (bacteria), or solely made of synthetic materials.



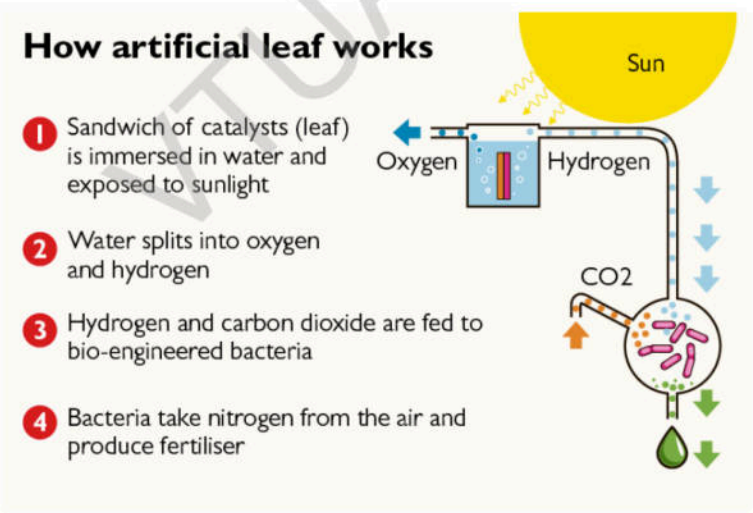

- The Bionic Leaf has the potential to be implemented in communities, such as urbanized areas to provide clean air as well as providing needed clean energy.
- The Bionic Leaf is an artificial leaf that interfaces a triple-junction *Si* wafer with amorphous silicon photovoltaic with hydrogen-and oxygen-evolving catalysts. These are

made from a ternary alloy, nickel-molybdenum-zinc *(NiMoZn)* and a cobalt–phosphate cluster (*Co-OEC*).

- The Co-OEC is able to operate in natural water at room temperature. Accordingly, the Bionic Leaf can be immersed in water and when held up to sunlight, it can effect direct solar energy conversion via water-splitting.
- *In natural photosynthesis, photosynthetic organisms produce energy-rich organic molecules from water and carbon dioxide by using solar radiation. Therefore, the process of photosynthesis removes carbon dioxide, a greenhouse gas, from the air.*




- *Artificial photosynthesis, as performed by the Bionic Leaf, is approximately 10 times more efficient than natural photosynthesis. Using a catalyst, the Bionic Leaf can remove excess carbon dioxide in the air and convert that to useful alcohol fuels, like isopropanol and isobutanol.*

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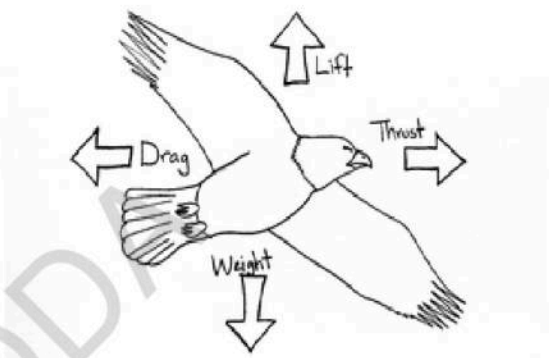
**Applications**

- Can be used for fertilizer production-new bionic leaf uses photovoltaic cells in conjunction with *Xanthobacter autotrophicus* bacteria to create a plastic called polyhydroxybutyrate (PHB) which can help for the sustainable production of ammonia and bio-fertilizers.

- The bacteria within the bionic leaf also help to remove carbon dioxide from the environment. Thus, bionic facades can be implemented in buildings to reduce energy consumption from cooling, absorb solar radiation, and reduce CO2 emissions.

**Bird flying**

- Bird flight is defined as the mode of locomotion used by birds while taking off and flying.
- This is observed to be one of the most complex forms of locomotion in the animal kingdom. Bird flying follows aerodynamic forces are used based on four forces: lift, thrust, drag, and gravity.




- By principle Thrust must equal Drag and Lift must equal to Gravity (Weight) in any straight and level flight.
- The x-section of a wing of a bird resembles a teardrop shape.
- When the leading edge of a streamlined wing cleaves the air, it pushes the air both up and down so that the amount of air passing above and below the wing should be and is the same. [Because the air molecules travel faster and over a longer distance, there is less pressure as per Bernoulli's Principle.] Thus the wing moves up into the area of lesser pressure (LIFT).
- But as air passes over the wing, friction is generated; the effect of this friction is to slow the wing and it is called DRAG. If the wing is tilted upward (increased angle of attack), the pressure on the upper part of the wing is reduced even further and lift increases; but so does the drag.
- The more the THRUST, the higher the angle of attack can be without stalling (loss of LIFT).
- The types of flight include two major types of modifications in their wings for gliding or soaring are found. The shape of a bird's tail also appears to be related to flight.
- Birds long range fly is expected to follow navigation. It is expected to follow

a) Nose Knows Theory- suggests that homing pigeons may use an olfactory map.

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b) Magnetic Map Theory- suggests that birds use the earth's magnetic field to obtain at least a partial map of its position. The earth's magnetic field becomes stronger as you travel away from the equator and toward the poles. In theory, a bird might be able to estimate its latitude based on the strength of the magnetic field.

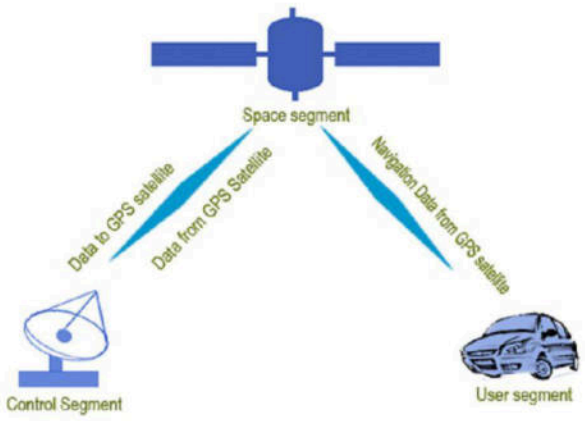
c) The use of compass-Sun, Moon and Stars.

d) The Sunset Clue and the Landmarks clue.

- Scientists have long known that the birds navigate using the earth's magnetic field. Now, a new study has found subtle mechanics in the brain of pigeons that allow them to find their way. This forms the basis to develop 2 important uses for mankind viz., GPS and aircrafts.

**GPS**

- A global positioning system (GPS) is a network of satellites and receiving devices used to determine the location of something on Earth. Some GPS receivers are so accurate they can establish their location within 1 centimeter.
- Roger L. Easton of the Naval Research Laboratory, Ivan A. Getting of the Aerospace Corporation, and Bradford Parkinson of the Applied Physics Laboratory invented GPS.
- The global positioning system (GPS) has three parts or elements. These components work together to make sure that the navigation, timing, and positioning information sent across are consistently accurate.




- GPS uses can be so listed as Location – determining a position, Navigation – getting from one location to another, Tracking – monitoring object or personal movement, Mapping – creating maps of the world.
- GPS parts are the space segment, control segment, and receivers.
- It works on the basic principle of exchange of radio waves between the ground stations, satellites, and the receivers. This transmission and reception of data prefer a trilateration (*the use of distances (or "ranges") for determining the unknown position coordinates of a point of interest, often around Earth (geo-positioning)*) mechanism of operation.

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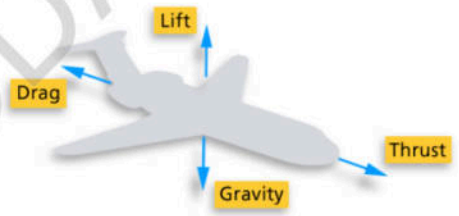
- During its working, an antenna, receives the signal, a receiver-processor unit, which converts the radio signal to a useable navigation solution; and a control/display unit, which displays the positioning information.
- 3 types of GPS are- Personal GPS, Commercial GPS and Military GPS.
- GPS also provides critical data for weather forecasting, agricultural planning, and even disaster relief efforts.

**Aircrafts**

- The aircraft flight is directly influenced by the principle of bird flight.
- Airplane wings are shaped to make air move faster over the top of the wing. When the air moves faster, the pressure of the air decreases. So the pressure on the top of the wing is less than the pressure on the bottom of the wing. The difference in pressure creates a force on the wing that lifts the wing up into the air.
- An aircraft in straight and level flight is acted upon by four forces: lift, gravity, thrust and drag. The opposing forces balance each other: Lift equals gravity, and thrust equals drag




a) *Thrust:* The force that moves an airplane forward through the air. Thrust is created by a propeller or a jet engine.
b) *Drag:* The air resistance that tends to slow the forward movement of an airplane.
c) *Gravity:* The force that pulls all objects towards the earth.
d) *Lift:* The upward force that is created by the movement of air above and below a wing. Air flows faster above the wing and slower below the wing, creating a difference in pressure that tends to keep an airplane flying.

**Lotus leaf effect (Super Hydrophobic and Self-Cleaning Surfaces)**

- The surface of a lotus leaf is an example of super hydrophobicity. On a super hydrophobic surface, the contact angle is greater than $150^{o}$, meaning almost no wetting of the surface by the liquid takes place.

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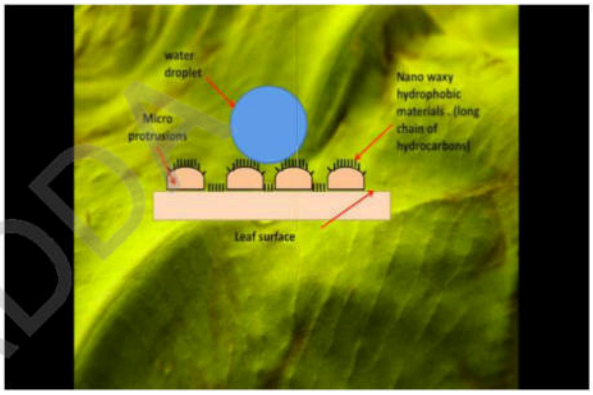
- This leads to the second property associated with lotus plants — the ability to stay spotlessly clean. As rain falls on a super hydrophobic surface like the lotus leaf, the water droplets roll easily off the leaf surfaces. As the droplets travel along the leaves, they pick up any dirt or other matter they encounter along the way. This process keeps the lotus leaves dry, clean and free of pathogens such as bacteria and fungi.

**Super hydrophobicity and Self-cleaning surfaces**

Development of self-cleaning surfaces is based on the concept of the 'Lotus effect'.

- Although many other plants have super hydrophobic surfaces with almost similar contact angles, the lotus shows better stability and perfection of its water repellency.
- This effect is observed due to the upper epidermis of the lotus leaf has developed some unrivaled optimizations.
- The extraordinary shape and the density of the papillae are the basis for the extremely reduced contact area between surface and water drops.




- The exceptional dense layer of very small epicuticular wax tubules is a result of their unique chemical composition. The mechanical robustness of the papillae and the wax tubules reduce damage and are the basis for the perfection and durability of the water repellency.
- A reason for the optimization, particularly of the upper side of the lotus leaf, can be deduced from the fact that the stomata are located in the upper epidermis. Here, the impact of rain and contamination is higher than on the lower epidermis. The lotus plant has successfully developed an excellent protection for this delicate epistomatic surface of its leaves.

Lotus effect has foresees a wide variety of beneficial to

- Textile applications in suiting, coats, furnishings, trousers, shower curtains, table covers etc. Fluorocarbon based fabrics are used in making sportswear, and industrial work wear.
- Development of paints that prevent corrosion of metallic materials.
- Roof tiles and various surface coatings.

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**Plant burrs (Velcro)**

- A bur (also spelled burr) is a seed or dry fruit or infructescence that has hooks or teeth. The main function of the bur is to spread the seeds of the bur plant, often through epizoochory.
- The bur of burdock was the inspiration for hook and loop fastener, also known as Velcro.

**Velcro**

- The fastener consisted of two components: a lineal fabric strip with tiny hooks that could 'mate' with another fabric strip with smaller loops, attaching temporarily, until pulled apart.
- Initially made of cotton, which proved impractical, the fastener was eventually constructed with nylon and polvester.

- Secure Outdoor Furniture, Equipment, and Décor. Velcro tape works well outdoors as long as it stays clean.
- Hang Kitchen Tools.
- Hang Photo Frames.
- Organize a Wardrobe.
- Fasten Cables Together.
- Hold a Rug or Mat in Place.
- Organize Garage Tools.

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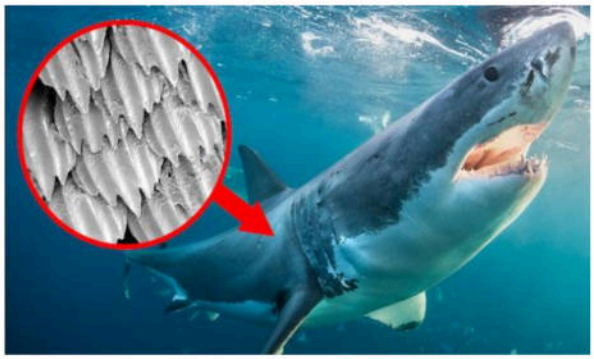
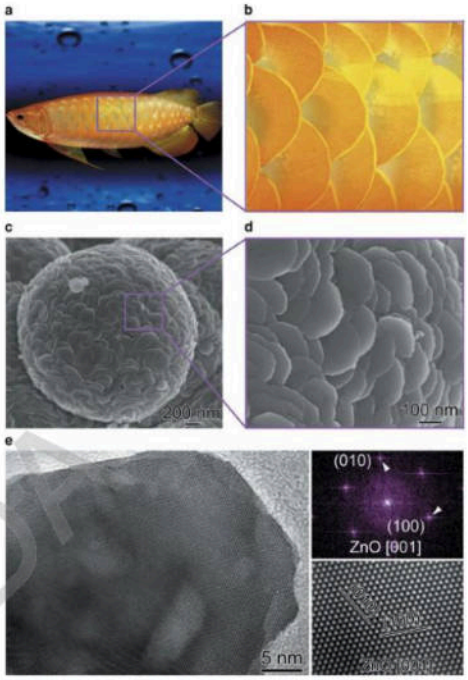
**Shark skin (Friction Reducing Swim Suits)**



- Shark skin covered with tooth-like scales structures known as denticles (tiny flat V-shaped scales) that make shark skin rough like sandpaper.
- For decades, scientists have suspected these denticles make sharks faster and more maneuverable by disrupting the flow of water over the fish and reducing any drag/and turbulence holding them back.

- It is found that the shark's skin, abundantly covered with placoid scales (dermal teeth), is a rough surface of grooves and ridges that form “nanostructured bulges” that allow sharks to swim with reduced resistance and prevent microorganisms from attaching to the body. This mechanism is made use of preparing swim suits.

**Swim Suits**




- During the designing of swim suits, people focus on basic component of fluid dynamics.
- That backward force imparted on our hand during swimming is called as is drag in action.
- One of the reasons that swimsuits are tight to the body is to help control the shape of the wearer's body.
- It effectively helps to make them more hydrodynamic. Added bio copying Shark skin, the modern and advanced swim suits are provided with denticles like structures.

**Kingfisher beak (Bullet Train)**

- Common Kingfishers measure 17 – 19 centimetres in length, weigh between 34 – 46 grams and have a wingspan of 25 centimetres. Their beak is around 4 centimetres long and pointed. Kingfisher Beak is also called as "bill”. The kingfishers have long, dagger-like bills. The bill is usually longer and more compressed in species that hunt fish, and shorter and broader in species that hunt prey off the ground.
- Once a kingfisher spies a fish (using special glare-reducing cells in its eyes), it leaves a perch and plunges into the water to grab it in its beak. Fish, however, have a defensive strategy that is hard to overcome. Specialized receptors along a fish’s body, known as a lateral line, sense disturbances in the flow of surrounding water. Any sudden movement of water—such as a compression wave from a diving bird—and fish are gone with a flick of the tail.
- But a kingfisher, hitting the water’s surface at over thirty-six feet (eleven meters) per second, manages to grab a fish in its beak before the fish detects it. This is possible because of the shape of the kingfisher’s beak.

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- A long and narrow cone, the kingfisher's beak parts and enters the water without creating a compression wave below the surface or a noisy splash above. The fine point of the conical beak presents little surface area or resistance to the water upon entry, and the evenly and gradually enlarging cross-section of the beak keeps fluid flowing smoothly around it as it penetrates further into the water column. This buys the bird crucial milliseconds to reach the fish before the fish knows to flee. The length of the beak is critical here: the longer it is, the more gradually the angle of the wedge expands. A shorter, fatter, or rounder beak would increase the wedge angle, resulting in a splash, a compression wave, and a fleeing fish.
- This kingfisher beak inspired a bullet train. In Japan, they had these very fast bullet trains. They were getting so fast that the typical bullet shape was causing a loud booming sound when these trains would exit typical train tunnels. The sound was waking up people who lived nearby. It disturbed the wildlife.
- The booming, it turned out, had to do with the shape of the face of the train. This was observed due to the cushion of air building up in front of that speeding train. To overcome the problem, when they designed front of the bullet train like the kingfisher's face it moved through without creating the boom and it saved 10-15% more energy because it was more aerodynamic. This concept was later developed to larger extent to have the present day bullet trains.

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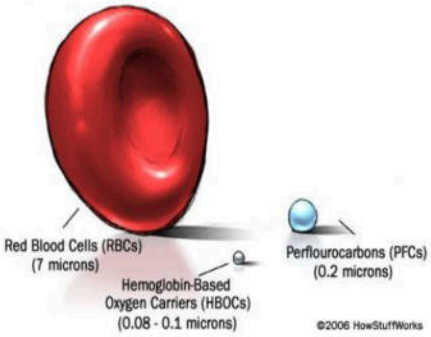
**Human Blood Substitutes**

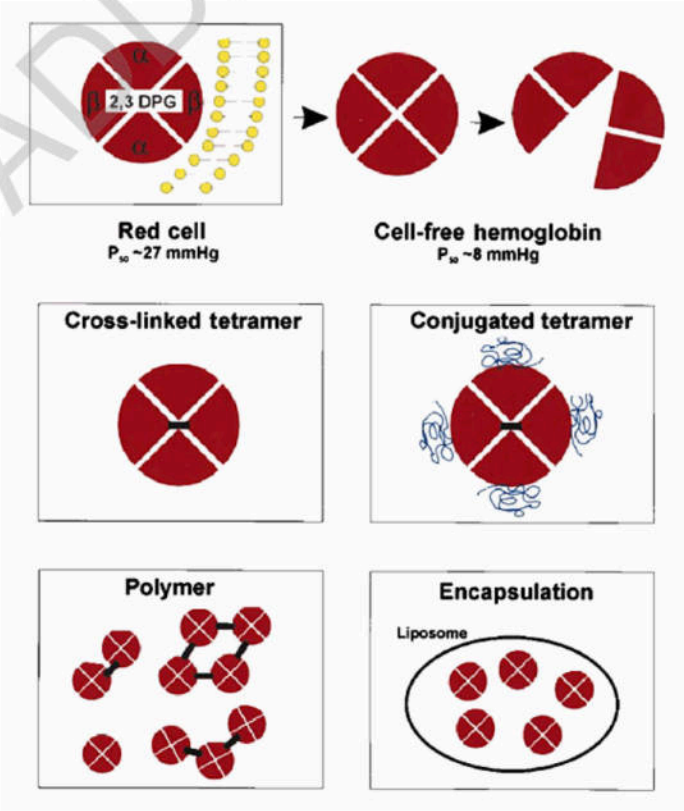
- Shortages in blood supplies and concerns about the safety of donated blood have fueled the development of so-called blood substitutes. The two major types of blood substitutes are volume expanders, which include solutions such as saline that are used to replace lost plasma volume, and oxygen therapeutics, which are agents designed to replace oxygen normally carried by hemoglobin in red blood cells.
- Of these two types of blood substitutes, the development of oxygen therapeutics has been the most challenging.

### (i) Hemoglobin-Based Oxygen Carriers (HBOCs)

- Oxygen therapeutics include agents called hemoglobin-based oxygen carriers (HBOCs), which are made by genetically or chemically engineering hemoglobin isolated from the red blood cells of humans or bovines.
- These dark red or burgundy colored blood substitutes are often made also from RBCs of expired human blood, cow blood, hemoglobin-producing genetically modified bacteria, or human placentas.




- HBOCs are semi-synthetic systems that utilize natural *Hb* as the oxygen-carrying component, either in chemically modified cell-free suspensions or conjugated and cross-linked with polymers along with protective enzymes, or encapsulated within microparticulate or nanoparticulate vehicles.
- Through a chemical process-polymerization, two or more three molecules bonded together to form a larger HBOC molecule but HBOCs are smaller than natural RBCs.
- In the case of outdated human or bovine RBCs, the *Hb* is isolated via cell lysis, purified by sterile filtration and chromatographic techniques and sterilized.
- HBOCs do not require refrigeration, are compatible with all blood types, and efficiently distribute oxygen to tissues. Using cell-free (acellular) *Hb* presents the advantage of minimum antigenicity and the ability to off-load oxygen in plasma, more efficiently because of the lack of interference by cell membrane.




- Examples of HBOCs include Hemopure, PolyHeme, Hemospan, Hemotech and Engineered Hemoglobin.

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***Advantages over human blood***

- Availability
- Long-term storage
- Lack of infectious risk.

***Primary concerns with HBOCs***

- Their potential to cause severe immune related reactions; a large number of them developed renal toxicity and cardiovascular complications.
- While natural RBCs remain in the bloodstream for about 100 days, HBOCs circulate in human blood for only a day.
- Side effects of HBOCs may include elevated blood pressure, abdominal discomfort, and a temporary reddish coloration of the eyes or skin.

**(ii) Perflourocarbons (PFCs)**

- Another group of agents developed and tested were perfluorocarbons, which effectively transport and deliver oxygen to tissues.
- These are totally synthetic artificial blood products derived from fluorine- and carbon-containing chemicals. They are chemically inert, but more effective than water or blood plasma in dissolving and absorbing oxygen in the lungs and then transporting oxygen throughout the body.
- PFCs remain in the bloodstream for about 48 hours. Because of their oxygen-dissolving ability, PFCs were the first group of artificial blood products studied by scientists. They are the first generation blood substitutes.
- Unlike the red colored HBOCs, PFCs are usually white. However, since they do not mix with blood they must be emulsified before they can be given to patients.
- PFCs are such good oxygen carriers that researchers are now trying to find out, whether it can reduce swollen brain tissue in traumatic brain injury.
- PFC particles may cause flu-like symptoms in some patients when they exhale these compounds, also cause complex side effects, including flulike reactions, and are not metabolized by the body.
- Examples of PFC blood substitutes are Fluosol-DA-20, Oxygent, Perftoran, Oxycyte and PHER-$O_2$.

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